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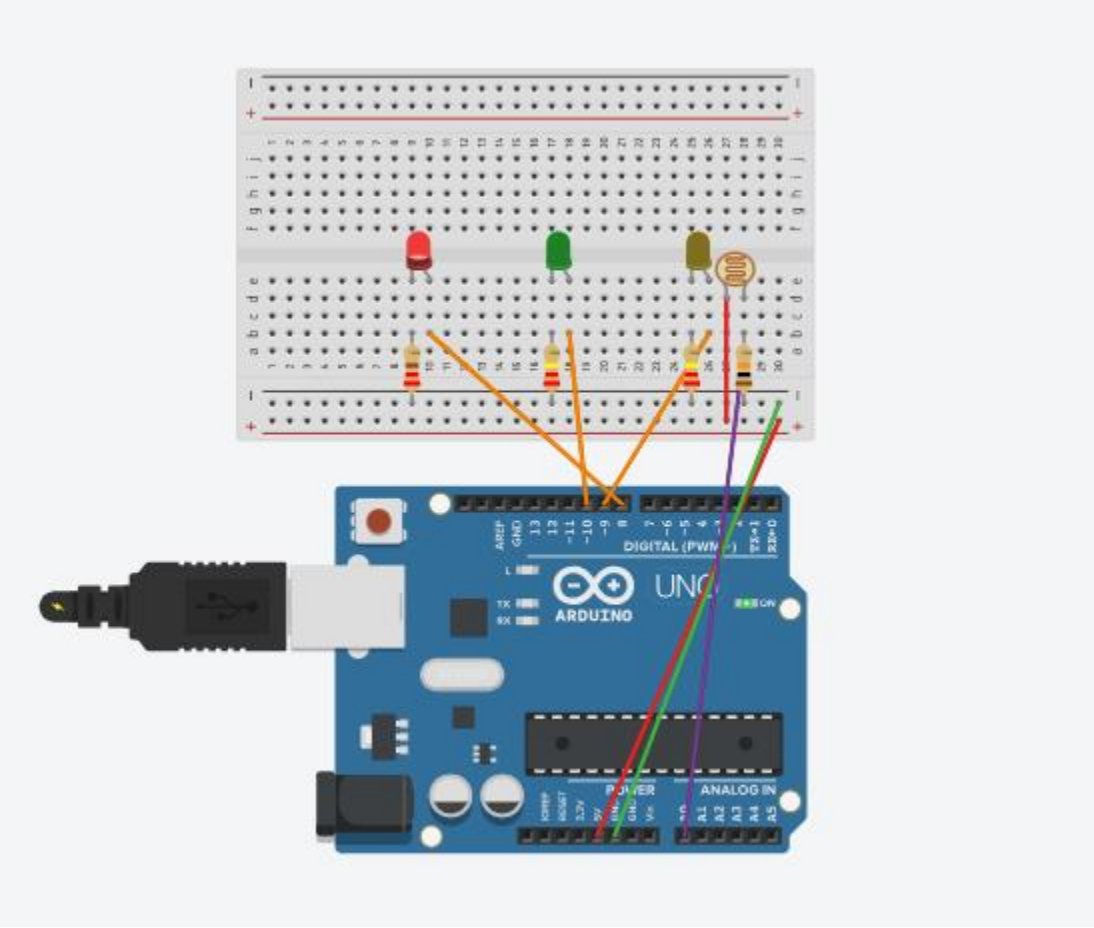
Semester: 1st
Program: MS Data Science

Part 1: Circuit Setup

1. Build an LDR voltage divider:- One leg of LDR → 5V - Second leg → A0 AND → 10kΩ resistor → GND

2. Connect LEDs: - Red → Pin 8- Yellow → Pin 9- Green → Pin 10

Each LED must include a 220Ω resistor to GND.



Part 2: Arduino Code

```
int ldrPin = A0;

int redLED = 8;

int yellowLED = 9;

int greenLED = 10;


void setup() {
  pinMode(redLED, OUTPUT);
```

```
  pinMode(yellowLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
}

void loop() {
  int sensorValue = analogRead(ldrPin);

  // Turn all LEDs OFF
  digitalWrite(redLED, LOW);
  digitalWrite(yellowLED, LOW);
  digitalWrite(greenLED, LOW);

  if (sensorValue < 300) {
    digitalWrite(redLED, HIGH);
  }
  else if (sensorValue < 700) {
    digitalWrite(yellowLED, HIGH);
  }
  else {
    digitalWrite(greenLED, HIGH);
```

```
  }


  delay(200);

}
```

Part 4: Reflection

1. The LDR (Light Dependent Resistor) changes its resistance based on the amount of light—low resistance in bright light and high resistance in darkness.
2. When the LDR is paired with a 10kΩ resistor in a voltage divider, the Arduino can read the changing voltage as an analog input.
3. Analog input gives a continuous value between 0 and 1023 instead of a simple ON/OFF reading, allowing detection of different light intensities.
4. Threshold values (300 and 700) were used to divide sensor readings into three levels: dark, medium light, and bright light.
5. During testing, the analogRead() values were observed in the Serial Monitor, and the thresholds were adjusted until LEDs responded correctly to each lighting condition.
6. This experiment demonstrated how physical changes (light) can be translated into electrical signals that a microcontroller can process.
7. LDR-based sensing is commonly used in IoT applications such as automatic streetlights, smart home lighting, energy-saving systems, and environmental monitoring.